

SST Non-Release Speculative Research Appendix v0.2

Composite Links, Phase Products, Modulo Classes, and Closed Modulation Ansätze

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Non-release status. This document records exploratory constructions, geometric intuitions, and unvalidated ansätze. Its contents are not part of the SST Canon, are not part of the release research-track companion, do not define SST predictions, and may be modified or rejected without requiring a Canon version change. No item in this appendix may be used to tune a Canon calibration, optimizer, null test, or stopping rule before formal promotion.

1 Epistemic ledger and promotion rule

Item	Non-release status
Periodic exponential modulation	[ANSATZ]; one closed modulation family among many.
Torus link $T(6, 9)$	[RESEARCH CANDIDATE]; exact three-component topology, no established SST selection law.
Three components times three phase labels	[STRUCTURAL ANSATZ]; naturally $\mathbb{Z}_3 \times \mathbb{Z}_3$, not automatically $\mathbb{Z}_9$.
Order-nine carry dynamics	[SPECULATIVE]; requires an explicit physical coupling law.
Doubling map modulo nine	[EXACT DISCRETE MATHEMATICS / PHYSICAL STATUS OPEN].
Rodin-style toroidal projection	[INSPIRATION ONLY]; geometry depends on representation choices.
Translation-induced trefoil flattening	[RETIRED]; not required by the action-phase mass-shell clock route.
Translation-induced stretching / uniform core thinning	[RETIRED]; no momentum-shape coupling has been derived.

Promotion into the release research track requires, at minimum:

1. independently defined physical variables and an evolution law;
2. dimensional consistency and conservation-law compatibility;
3. comparison against alternative moduli, geometries, and phase laws;
4. a preregistered falsifier and a minimal analytic or numerical test;
5. no use of the candidate pattern in calibration or model selection;
6. reproducible improvement over simpler control models.

2 Retired translation-induced shape hypotheses

The exploratory suggestions that a translating electron trefoil must flatten, stretch, or uniformly thin as its speed approaches c are retired. The v0.8.28 release route instead uses an internal action–phase Hamiltonian,

$$H(P, I) = \sqrt{P^2 c^2 + E_0^2(I)}, \quad (1)$$

which produces the conditional phase-rate law

$$\dot{\theta} = \Omega_0 \frac{E_0}{H} = \frac{\Omega_0}{\gamma} \quad (2)$$

without introducing a shape variable. If shape coordinates q are added only through $E_0(I, q)$, then

$$\partial_q H = \frac{E_0}{H} \partial_q E_0, \quad (3)$$

so the rest-shape extrema are unchanged at finite momentum. A future translation-dependent deformation may be reconsidered only if an independently derived coupling $H_{Pq} \neq 0$ is found. The retired hypotheses may not be used in calibration, particle-size estimates, or relativistic predictions.

3 Periodic exponential modulation

A bounded periodic radius modulation may be written as

$$\rho(u) = \rho_0 \exp[\eta \sin(nu)], \quad n \in \mathbb{Z}, \quad (4)$$

with a torus-knot or torus-link parameterization

$$\mathbf{X}_{p,q,n,\eta}(u) = \begin{pmatrix} [R + \rho(u) \cos(qu)] \cos(pu) \\ [R + \rho(u) \cos(qu)] \sin(pu) \\ \rho(u) \sin(qu) \end{pmatrix}, \quad 0 \leq u < 2\pi. \quad (5)$$

For integer p, q, n , the parameterization is closed. Closure alone does not establish preservation of knot type, finite reach, adequate core clearance, or dynamical stability. The parameter η must remain free in the primary test. Assignments such as $\eta = \ln \phi$ are secondary hypotheses and are not permitted in an optimizer before the free- η comparison.

The mandatory controls include an unmodulated embedding, ordinary sinusoidal or Fourier modulation, a spline family with comparable degrees of freedom, and equalized length or energy constraints. The exponential family is promoted only if it produces a reproducible dynamical advantage not explained by parameter count or geometric clearance.

4 The $T(6, 9)$ three-trefoil candidate

For the torus link $T(p, q)$, the number of components is

$$d = \gcd(p, q). \quad (6)$$

Consequently,

$$\gcd(6, 9) = 3, \quad (7)$$

and each component has reduced torus-knot type

$$T\left(\frac{6}{3}, \frac{9}{3}\right) = T(2, 3), \quad (8)$$

so $T(6, 9)$ consists of three trefoil components. With consistent orientation, the pairwise linking number of parallel torus-link components is

$$\text{Lk}_{ij} = \frac{pq}{d^2} = 6. \quad (9)$$

These are topological facts. They do not imply that $T(6, 9)$ is selected by Euler or Biot–Savart dynamics, that it is more stable than alternative three-component links, or that it realizes a particle state.

Hypothesis 4.1 (Composite-link candidate). *After a single resolved trefoil reaches at least the declared numerical KAM–3 gate under a fixed protocol, $T(6, 9)$ may be compared with control three-component links under matched core radius, circulation, total length, helicity metadata, and minimum clearance.*

5 Three-by-three phase labels

Label the three components by $a \in \mathbb{Z}_3$ and three local phase sectors by $k \in \mathbb{Z}_3$. The nine labels form

$$(a, k) \in \mathbb{Z}_3^{\text{component}} \times \mathbb{Z}_3^{\text{phase}}, \quad |\mathbb{Z}_3 \times \mathbb{Z}_3| = 9. \quad (10)$$

This product group is not cyclic:

$$\mathbb{Z}_3 \times \mathbb{Z}_3 \not\cong \mathbb{Z}_9, \quad (11)$$

because every nonzero element of $\mathbb{Z}_3 \times \mathbb{Z}_3$ has order three, whereas $\mathbb{Z}_9$ contains elements of order nine.

A genuine order-nine orbit therefore requires an additional successor law. One possible bookkeeping map is the base-three carry operation

$$S(a, k) = \begin{cases} (a + 1, k), & a = 0, 1, \\ (0, k + 1 \pmod{3}), & a = 2, \end{cases} \quad S^9 = \text{id}. \quad (12)$$

Equation (12) is only a discrete indexing rule until a physical interaction derives it.

6 Doubling modulo nine

The map

$$x_{m+1} = 2x_m \pmod{9} \quad (13)$$

is an exact finite dynamical system. Its orbit decomposition is

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 7 \rightarrow 5 \rightarrow 1, \quad (14)$$

$$3 \leftrightarrow 6, \quad (15)$$

$$0 \rightarrow 0. \quad (16)$$

The six-cycle is the multiplicative unit group

$$\mathbb{Z}_9^\times = \{1, 2, 4, 5, 7, 8\}, \quad \text{ord}_9(2) = 6. \quad (17)$$

This is rigorous number theory. The familiar decimal digital-root representation follows from $10 \equiv 1 \pmod{9}$ and is therefore partly tied to the choice of base ten. A physical role for Eq. (13) requires an independently derived nine-state variable and a reason that one physical update acts as multiplication by two modulo nine.

7 Separation from the KAM release programme

The v0.8.25 KAM release programme tests relative equilibria, reduced Floquet ellipticity, Krein signatures, nonlinear twist, finite-order non-resonance, Poincaré maps, action drift, and invariant-torus survival. The constructions in this appendix may not be used to preselect a stable state. The valid order is

unbiased KAM search $\longrightarrow$ observed structure $\longrightarrow$ post-hoc comparison with speculative candidates. (18)

A candidate is rejected when its apparent advantage disappears under matched controls, resolution refinement, alternative parameterizations, or removal of base-dependent encoding choices.

Child-level analogy. The appendix is a sketchbook, not the construction manual. A beautiful gear pattern can suggest what to test, but the machine must still work when the pattern is hidden from the engineer.

References

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